# Practical No. 8: Correlation Coefficient and Rank Correlation Coefficient

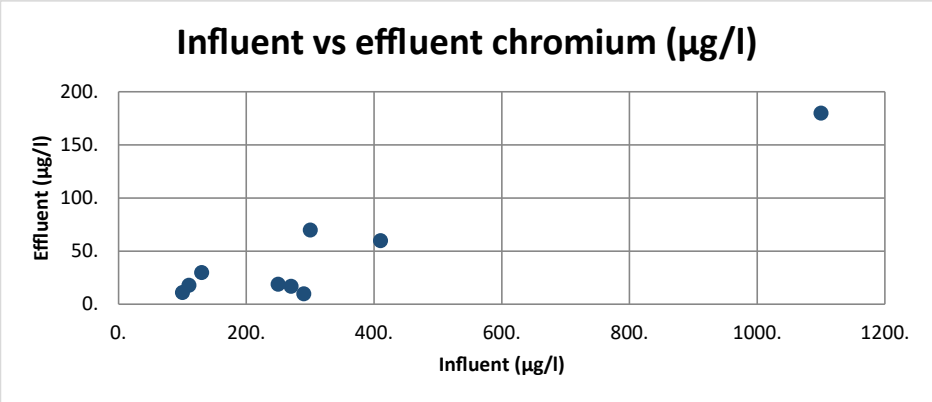
Blue = data from question; green = results. $r = (n\Sigma xy - \Sigma x\Sigma y)/\sqrt{[(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)]}$; $\rho = 1 - 6\Sigma d^2/[n(n^2 - 1)]$. Σ in column L.

| Ex.1 Correlation between sales and expenses of 10 firms (₹ '000) | | | | | | | | | | | Σ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sales (x) | 50. | 50. | 55. | 60. | 65. | 65. | 65. | 60. | 60. | 50. | 580. |
| Expenses (y) | 11. | 13. | 14. | 16. | 16. | 15. | 15. | 14. | 13. | 13. | 140. |
| x·y | 550 | 650 | 770 | 960 | 1040 | 975 | 975 | 840 | 780 | 650 | 8190 |
| $x^2$ | 2500 | 2500 | 3025 | 3600 | 4225 | 4225 | 4225 | 3600 | 3600 | 2500 | 34000 |
| $y^2$ | 121 | 169 | 196 | 256 | 256 | 225 | 225 | 196 | 169 | 169 | 1982 |
| Results | n | 10 | r | 0.7866 | CORREL | 0.7866 | $r^2$ | 0.6187 | | | |

Expected: r = 0.7866

| Ex.2 Chromium (µg/l) in influent (x) and effluent (y): scatter plot and correlation coefficient | | | | | | | | | | | Σ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Influent (x) | 250. | 290. | 270. | 100. | 300. | 410. | 110. | 130. | 1100. | | 2960. |
| Effluent (y) | 19. | 10. | 17. | 11. | 70. | 60. | 18. | 30. | 180. | | 415. |
| x·y | 4750 | 2900 | 4590 | 1100 | 21000 | 24600 | 1980 | 3900 | 198000 | | 262820 |
| $x^2$ | 62500 | 84100 | 72900 | 10000 | 90000 | 168100 | 12100 | 16900 | 1210000 | | 1726600 |
| $y^2$ | 361 | 100 | 289 | 121 | 4900 | 3600 | 324 | 900 | 32400 | | 42995 |
| Results | n | 9 | r | 0.9425 | CORREL | 0.9425 | $r^2$ | 0.8882 | | | |

Expected: r = 0.9424




| Ex.3 Rank correlation coefficient for rankings of 10 elements | | | | | | | | | | | Σ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Element | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| x (rank) | 10 | 6 | 7 | 3 | 4 | 2 | 8 | 5 | 1 | 9 | |
| y (rank) | 8 | 4 | 10 | 2 | 5 | 7 | 6 | 3 | 1 | 9 | |
| d = x − y | 2 | 2 | -3 | 1 | -1 | -5 | 2 | 2 | 0 | 0 | |
| $d^2$ | 4 | 4 | 9 | 1 | 1 | 25 | 4 | 4 | 0 | 0 | 52 |
| Results | n | 10 | $\Sigma d^2$ | 52 | ρ | 0.6848 | CORREL | 0.6848 | | | |

Expected: ρ = 0.6848

| Ex.4 Marks in selection test (x) and proficiency test (y): rank correlation coefficient | | | | | | | | | | | Σ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| x (marks) | 10 | 15 | 12 | 17 | 13 | 16 | 24 | 14 | 22 | 20 | |
| y (marks) | 30 | 42 | 45 | 46 | 33 | 34 | 40 | 35 | 39 | 38 | |
| Rank of x | 10. | 6. | 9. | 4. | 8. | 5. | 1. | 7. | 2. | 3. | |
| Rank of y | 10. | 3. | 2. | 1. | 9. | 8. | 4. | 7. | 5. | 6. | |
| d | 0. | 3. | 7. | 3. | -1. | -3. | -3. | 0. | -3. | -3. | |
| $d^2$ | 0. | 9. | 49. | 9. | 1. | 9. | 9. | 0. | 9. | 9. | 104. |
| Results | n | 10 | $\Sigma d^2$ | 104 | ρ | 0.3697 | CORREL | 0.3697 | | | |

Expected: ρ = 0.3697 (no tied values, so no tie correction needed)